

Muhammed Aksoy

publishes as Mohamed Elsaedy

Machine Learning Research Engineer — SAR/EO-IR Computer Vision, Open-World Recognition & Model Reliability

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Google Scholar (4 papers, 19 citations, h-index 3) · ORCID 0000-0002-6225-7915 · Istanbul, Turkey · Open to U.S. relocation

Summary

Machine learning research engineer and Ph.D. candidate with 5 years of applied computer-vision research on multi-modal and radar imagery. Published first author on IEEE cross-spectral image registration (infrared-to-optical cGAN translation and Swin Transformer Siamese matching) with measured gains of 17% nn-mAP and 24% homography correctness over the prior state of the art, delivered under a funded TUBITAK national research project. Currently leads open-world SAR automatic target recognition research that separates confident recognition from principled rejection of unknown targets. Builds the full stack the work needs: dataset protocols, reproducible training pipelines, calibrated evaluation, edge deployment on NVIDIA Jetson, and Python/FastAPI services.

Technical Skills

Languages: Python, C++, MATLAB, SQL, Bash, TypeScript

ML / Deep Learning: PyTorch, scikit-learn, OpenCV, NumPy, transformers (Swin/SwinV2, ViT), CNNs, GANs (pix2pix/cGAN, PatchGAN), Siamese and metric learning, few-shot learning, self-supervised keypoint learning

Computer Vision: Multi-modal and cross-spectral image registration, keypoint detection and description, image-to-image translation, object detection, image fusion, homography estimation, SAR ATR, remote sensing

Model Reliability: Open-set and open-world recognition, out-of-distribution detection, uncertainty quantification, conformal and calibrated risk control, explainability auditing, domain-shift and degradation testing

Deployment / Edge: CUDA, TensorRT, ONNX, Docker, NVIDIA Jetson Nano / Xavier NX / AGX Orin, model quantization and pruning, real-time inference optimization

Backend & Infrastructure: FastAPI, Celery, Redis, PostgreSQL, REST APIs, React/Vite, Git, Linux, structured logging, test automation, CI workflows

Research Engineering: Config-driven experiment runners, dataset manifests and versioning, multi-seed reproducibility protocols, ablation design, leakage control, benchmark reproduction, LaTeX

Datasets & Benchmarks: MSTAR, FUSAR-Ship, SLICY, VEDAI, ImagePair, LSST-like survey data (FITS/SNANA)

Experience

Research Engineer / Doctoral Researcher

Oct 2021 – Present

Istanbul Medipol University — Computer Vision & AI Lab

Istanbul, Turkey

- Delivered the cross-spectral registration work package of TUBITAK national research project 118E891, producing two first-author IEEE conference papers (2022, 2023) that have since drawn 14 citations.
- Built an infrared-to-optical image translation pipeline (pix2pix conditional GAN, U-Net-256 generator, PixelGAN discriminator) trained on 9,001 paired 512×512 aerial images; cut FID from 121.7 to 107.0 and, by retraining the MultiPoint matcher on translated imagery, raised nearest-neighbor mAP from 0.446 to 0.524 and matching score from 0.187 to 0.198 over the published baseline.
- Replaced the CNN encoder of the MultiPoint keypoint network with a Swin Transformer V2 Siamese encoder trained from scratch with a cosine-similarity descriptor alignment loss; on the VEDAI thermal/optical benchmark this improved nn-mAP by 17.3%, matching score by 8.4%, homography correctness by 24%, and repeatability by 1.5% against the MultiPoint baseline.
- Leads doctoral research on NeuroATR, an open-world few-shot SAR target recognition framework using SwinV2 backbones and calibrated knownness scoring, able to reject unknown targets rather than force-classify them — the failure mode that makes closed-set ATR unusable in the field.
- Designed the evaluation protocol that makes those results defensible: exact-match MSTAR-7 reporting, strict known-only multi-seed checkpointing with leakage control, SLICY and clutter rejection diagnostics, T-72 configuration-variant near-OOD tests, extended-operating-condition depression-angle stress, and FUSAR-Ship external unknown transfer.
- Developed a tri-source reliability layer decomposing recognition confidence into logit-energy, prototype-distance, and structural-disagreement evidence, with retrain-level ablations, generic OOD baselines, calibration diagnostics, and latency measurement for deployment budgeting.
- Screened and extracted metrics from 1,000+ papers to build a source-verified systematic review of deep learning for multi-source multi-modal image registration, including a reproducible Python screening, deduplication, and traceability pipeline across medical, remote-sensing, and cross-spectral domains.
- Optimized real-time object detection models for NVIDIA Jetson deployment (TensorRT/ONNX) and contributed to thermal-optical fusion and multi-modal registration projects for aerial platforms.
- Teaching assistant and course coordinator for Machine Learning, Deep Learning, Computer Vision, and Embedded AI; mentored graduation-project teams through problem definition, evaluation, and defense.

R&D Intern, TUTEL Lab

TUBITAK BILGEM (Turkish national defense & IT research center)

Jun 2021 – Aug 2021

Kocaeli, Turkey

- Built an automatic license plate recognition subsystem for a national intelligent transportation program, covering plate detection, character segmentation, and recognition.
- Implemented FPGA-based image processing blocks (Vivado) for real-time video pipelines.

Machine Learning Researcher

Kafein Technology Solutions

Sep 2020 – Oct 2020

Istanbul, Turkey

- Developed Parkinson's disease classification models from wearable accelerometer time series collected across Turkish hospitals, covering signal preprocessing, feature extraction, and patient-wise validation.

Engineering Intern, Computer Vision Lab

Istanbul Medipol University

Jul 2020 – Sep 2020

Istanbul, Turkey

- Implemented neural networks from scratch and optimized detection models for high-speed inference on Jetson Nano and Xavier NX, working on pruning, quantization, and TensorRT conversion.

Publications

Peer-reviewed (published under the name Mohamed Elsaeidy)

1. **M. Elsaeidy**, M. E. Erkol, B. K. Gunturk, H. F. Ates, "Infrared-to-Optical Image Translation for Keypoint-Based Image Registration," *IEEE Signal Processing and Communications Applications Conference (SIU)*, 2022. (11 citations)
2. **M. Elsaeidy**, I. C. Yagmur, H. F. Ates, B. K. Gunturk, "Swin Transformer-based Siamese Network for Thermal and Optical Image Registration," *IEEE Signal Processing and Communications Applications Conference (SIU)*, 2023. (3 citations)
3. M. Z. Alam, S. Kelowani, **M. Elsaeidy**, "Spatio-Angular Resolution Trade-off in Face Recognition," *International Conference on Emerging Trends and Applications in AI (ICETAI)*, 2023; extended preprint arXiv:2402.07263, 2024. (5 citations)

Under review / in preparation

4. **M. Aksoy**, I. Demirkiran, B. K. Gunturk, "NeuroATR: Classifier-Centered Open-World SAR ATR with Broad Diagnostic Unknown-Target Evaluation," *submitted to ISPRS Journal of Photogrammetry and Remote Sensing*, 2026.
5. **M. Aksoy**, B. K. Gunturk, "Deep Learning for Multi-Source Multi-modal Image Registration: A Systematic Evidence Map and Quantitative Review," in preparation.
6. **M. Aksoy**, B. K. Gunturk, "Evidence-Source Complementarity for Open-World SAR Target Recognition," in preparation for *IEEE TGRS*.
7. A. A. Boudjelal, **M. Aksoy**, H. Arslan, "Neural-Assisted Parametric Path Estimation for Fractional Delay-Doppler Channels," in preparation.

Selected Projects

- **Cross-Spectral Registration Suite (M.Sc. thesis, TUBITAK 118E891)**. Two complementary approaches to the same problem — close the appearance gap generatively (pix2pix cGAN) or learn a shared representation directly (SwinV2 Siamese encoder with cosine descriptor alignment). Both beat the MultiPoint/SuperPoint baselines on ImagePair and VEDAI; results published at IEEE SIU 2022 and 2023.
- **NeuroATR — Open-World SAR ATR Platform**. Config-driven research platform for few-shot SAR recognition with energy-based uncertainty and prototype-structured representations, built for strict reproducibility (dataset manifests, seed control, leakage-audited checkpointing) and external unknown-target stress testing on single-GPU hardware.
- **OpenManus Agent Runtime**. Extended an open-source autonomous LLM-agent runtime with sandboxed code execution, live tool observability, MCP server integration, and long-running task orchestration across a Python/FastAPI/PostgreSQL/React stack.
- **Risk-Controlled Astronomical Transient Triage**. Rare-transient detection pipeline for LSST-like surveys over 1.5M+ objects, using conformal calibration to hold a target error rate under degradation stress tests — the same reliability question as ATR, in a different domain.

Education

Ph.D., Computer Engineering — Istanbul Medipol University

2023 – expected 2027

Dissertation: reliable open-world SAR automatic target recognition under domain shift.

M.Sc., Computer Engineering — Istanbul Medipol University, GPA 3.7/4.0

2021 – 2023

Thesis: deep learning for infrared-optical image registration (basis of both IEEE SIU papers).

B.Sc., Electrical & Electronics Engineering — Istanbul Medipol University, GPA 3.13/4.0

2019 – 2021

Computer Software Engineering (transferred) — University of Science and Technology, GPA 3.58/4.0

2016 – 2019

Additional

Funded research: Contributor, TUBITAK project 118E891 (multi-modal image registration)

Teaching: TA and course coordinator — Machine Learning, Deep Learning, Computer Vision, Embedded AI / Image Processing; graduation-project mentoring and evaluation

Languages: Arabic (native), English (professional working), Turkish (professional working)